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Modern engineering increasingly depends on computational models that must remain reliable under stochastic variability, sparse or noisy measurements, incomplete physical information, and the high cost of repeated high-fidelity simulation. These challenges are especially important in manufacturing, which contributed approximately \$2.4 trillion to U.S. GDP in 2023.<sup>[1]</sup> They also motivate current research priorities in advanced manufacturing, uncertainty quantification, inverse methods, and scientific artificial intelligence at agencies such as the National Science Foundation (NSF) and the U.S. Department of Energy (DOE).<sup>[5]</sup> My research asks a central question: *how can we infer hidden physical quantities, discover governing structure, and accelerate engineering prediction while rigorously representing stochasticity and uncertainty?* I address this question through Bayesian inverse modeling, stochastic physics discovery, causal reasoning, and scientific machine learning, with applications in manufacturing, materials, thermal, semiconductor, and other engineered systems.

## **I. Summary of previous contributions**

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### **Uncertainty-aware inverse modeling**

Inverse problems arise when quantities needed for engineering decisions cannot be measured directly. My work on inverse heat conduction developed a Bayesian spatiotemporal framework that combines heat-transfer physics with statistical inference to reconstruct unobserved thermal quantities from sparse measurements while quantifying uncertainty. The work first appeared as an ASME conference paper in 2023 and was expanded into the *ASME Journal of Heat and Mass Transfer* in 2024.<sup>[2]</sup> I extended this principle to instrumented indentation, where stochastic data-consistent inversion is used to recover distributions of elastoplastic properties from load-displacement information while accounting for measurement noise and manufacturing-induced variability. The current journal extension is being developed with collaborators at NIST and Oak Ridge National Laboratory. Together, these studies establish a foundation for recovering latent physical quantities and property distributions from indirect engineering measurements.

### **Physics discovery under uncertainty**

Parameter estimation is insufficient when the assumed governing model is itself incomplete. To address this problem, I developed the stochastic inverse physics-discovery framework published in *Communications Physics* in 2026. The method represents unknown coefficients in candidate governing equations probabilistically and identifies distributions that are consistent with observed system behavior. Across ten scenarios spanning nonlinear oscillators, chaotic systems, ecological data, and infiltration dynamics, the framework reduced coefficient error by about 82% relative to comparison methods and achieved exact structural identification in seven of ten cases.<sup>[3]</sup> This work reframes governing-equation discovery as an uncertainty-aware inverse problem rather than a noise-sensitive curve-fitting task. My complementary work on counterfactual explanations investigates how causal evidence can distinguish dynamically meaningful interactions from spurious candidate terms.

### **Scientific machine learning for efficient computation**

Many engineering workflows remain too slow for repeated inference and optimization because each decision requires another high-fidelity simulation. My CRONet work addresses this bottleneck using a convolutional recurrent operator-approximator network for topology optimization. On benchmark problems, CRONet reduced computational time by up to 78% while maintaining compliance predictions within 5% of conventional finite-element solutions.<sup>[4]</sup> The work was later extended through collaborative hardware acceleration on AMD Versal AIE-ML engines. These contributions establish a pathway for incorporating fast learned operators into repeated design, inverse, and decision workflows without treating computational speed as a substitute for physical reliability.

## II. Future research and opportunities

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My future research will connect these contributions into adaptive engineering models that can update from new data, recognize when their assumptions are inadequate, and determine what measurement, simulation, or experiment should come next. The initial research program will focus on three closely connected directions.

### **Uncertainty-aware inverse modeling for autonomous characterization**

I will develop probabilistic inverse methods that distinguish intrinsic physical variability from measurement noise, numerical approximation, surrogate error, and model discrepancy rather than collapsing these effects into a single uncertainty term. Hierarchical Bayesian models, data-consistent inversion, probabilistic surrogates, and sequential inference will provide the methodological foundation. Autonomous material characterization will serve as an early testbed, where indentation, thermal, image, and other sensing modalities can be combined with physics-based simulation to infer spatially varying material properties and their uncertainty. The same framework can support thermal processes, process monitoring, and semiconductor metrology. This research aligns directly with DOE Advanced Scientific Computing Research priorities in uncertainty quantification and inverse methods and with NSF CMMI's Advanced Manufacturing, Design, and Operations program, which supports foundational research in intelligent and data-enabled manufacturing systems.<sup>[5]</sup>

### **Physics and causal discovery for self-correcting models**

A second direction will address systems in which the governing structure itself is uncertain. I will extend stochastic physics discovery to partially known models, heterogeneous observations, and changing operating conditions, while using counterfactual explanations and complementary causal diagnostics to test whether recovered interactions remain stable under perturbation and transferable across experiments. The longer-term objective is a self-correcting modeling loop in which parameter updating first attempts to explain new observations; persistent model discrepancy then triggers discovery of missing governing terms, followed by model recalibration and targeted experimentation. In materials and manufacturing, this creates a pathway toward closed-loop discovery of process-property and constitutive relationships. The agenda is especially well aligned with NSF's Designing Materials to Revolutionize and Engineer our Future (DMREF) program, which emphasizes iterative integration of theory, computation, data, and experiment and currently supports projects of approximately \$1.5–2.0 million over four years, as well as DOE's Advanced Materials and Manufacturing Technologies Office.<sup>[6]</sup>

### **Trustworthy scientific machine learning for digital engineering**

Building on CRONet, I will develop fast surrogate and operator-learning models for three-dimensional and multiphysics engineering problems while incorporating uncertainty calibration and diagnostics for predictions outside the reliable operating regime. Applications will include topology optimization, manufacturing process models, image-based geometric inspection, and other digital engineering systems. An ongoing image-based inspection project combining multi-view reconstruction with controlled-illumination surface characterization provides one experimental pathway from sensing to quality decisions. The goal is not simply to replace simulation with machine learning, but to develop models that indicate when a prediction can be trusted, when additional high-fidelity computation is required, and when new physical data should be collected.

These research directions form a coherent program in which inverse modeling recovers hidden states and properties, physics and causal discovery identify missing structure, and scientific machine learning makes repeated inference and decision making computationally practical. They provide natural opportunities for early-career proposals through NSF CAREER and the DOE Early Career Research Program, followed by larger collaborative efforts through NSF CMMI, DMREF, DOE programs, national laboratories, and industrial partners.<sup>[5][6]</sup> My existing collaborations with NIST and Oak Ridge National Laboratory provide a foundation for external validation and interdisciplinary partnerships. I plan to build a research group centered on reproducible software, documented benchmarks, FAIR data practices, and progressive student ownership of methodological questions. Ultimately, my goal is to develop engineering models that remain useful not because they are assumed to be correct, but because they can quantify uncertainty, recognize when their assumptions fail, and update their predictions and decisions accordingly.

**Selected references.** [1] NIST, [U.S. Manufacturing Economy](#). [2] Olabiyi et al., [ASME J. Heat Mass Transfer](#), 2024. [3] Olabiyi, Hu, and Iquebal, [Communications Physics](#), 2026. [4] Olabiyi, Yang, and Iquebal, [Manufacturing Letters](#), 2025. [5] NSF, [Advanced Manufacturing, Design, and Operations](#); DOE ASCR, [Applied Mathematics](#). [6] NSF, [DMREF](#); DOE, [AMMTO](#).